Collisional spin-coherence battery on a non-collinear thermoelectric spin valve

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We study a Coulomb-blockaded quantum dot coupled to non-collinear magnetic leads and to a stream of ancilla qubits. The nonequilibrium leads generate spin coherence in the singly occupied sector. A temperature difference and a load voltage place the device in a thermoelectric engine regime with positive electrical output power. Matched-gap exchange collisions then export part of the spin coherence into the ergotropy of the outgoing ancillas, opening a second power channel. Using lead-resolved currents built from the same dissipators that generate the dynamics, we compute cycle-averaged electrical power and battery charging power at the stroboscopic fixed point. A control hierarchy—collinear leads, pure dephasing, incoherent exchange, and matched populations—shows that the battery channel requires coherent exchange, while the engine branch survives without it. The collisional export map is summarized by a short companion analysis of impedance-matched harvesting. The construction is a sequential-tunneling scaffold, not a NEGF theory, and coherent ergotropy remains an upper bound without a phase reference.

I. INTRODUCTION

Quantum-dot thermoelectric devices convert heat into electrical work at the nanoscale [1–4]. When the leads are magnetic and non-collinear, the same nonequilibrium drive that produces charge current also polarizes the dot spin away from the energy basis, so a stationary spin coherence appears in the singly occupied sector [5]. Separately, quantum batteries and repeated-interaction models ask how fast a nonpassive state can be prepared and charged [6–11].

Our question is simple. *Can a non-collinear thermoelectric spin valve keep a positive electrical output while a collisional tape harvests its spin coherence as battery ergotropy?* If yes, what controls separate the two channels?

We answer this in a minimal Coulomb-blockade model with sequential tunneling to polarized leads and resonant exchange collisions with ground-state ancillas. Three results structure the paper. (i) In a near-degenerate window, non-collinearity produces a usable spin coherence, while collinear leads and large level splitting suppress it. (ii) A temperature bias together with a load voltage places the device on a thermoelectric engine branch with $P_{\mathrm{el}} > 0$. (iii) Coherent exchange collisions open a second channel $P_{\mathrm{erg}} = r\mathcal{W}_A$, which vanishes under incoherent exchange, collinear leads, or no collision.

We do not claim that exporting coherence restores a higher collinear electrical power. We treat electrical and battery powers as coexisting outputs. We report total ancilla ergotropy as an upper bound: without a phase reference the coherent part is not free under energy-preserving discharge [9, 12].

II. MODEL

A. Dot and leads

The strong-blockade Fock space is $\{|0\rangle, |\uparrow\rangle, |\downarrow\rangle\}$ with

$$H_D = \varepsilon(n_\uparrow + n_\downarrow) + \frac{\Delta}{2}(n_\uparrow - n_\downarrow). \tag{1}$$

Each lead $\alpha \in \{L, R\}$ has broadening matrix Γ_α with polarization p_α and, for the right lead, tilt (θ_R, ϕ_R). Temperatures $T_L > T_R$ and chemical potentials μ_L, μ_R define the bias $V = \mu_L - \mu_R$. Between collisions the reduced density matrix evolves under a sequential-tunneling Liouvillian built from spinor jump operators of Γ_α (Appendix A). The generator is completely positive by con-

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struction. It is *not* a derived Redfield equation; all claims below are inside this scaffold.

B. Collisions

A second, permanently occupied ancilla qubit with gap $\Delta_A = \Delta$ interacts for a time τ through

$$H_{\text{int}} = \hbar g\big(|{\uparrow_D}{\downarrow_A}\rangle\langle{\downarrow_D}{\uparrow_A}| + \text{h.c.}\big), \tag{2}$$

so $\theta = g\tau$. On the active block the free Hamiltonian is resonant and $[H_D + H_A, H_{\text{int}}] = 0$. Fresh ancillas arrive with period T and excited population α (usually $\alpha = 0$). The stroboscopic map is waiting under the leads, then one collision. Observables are evaluated at the periodic fixed point.

C. Bookkeeping

Lead-tagged dissipators define

$$J_{N,\alpha} = \text{Re Tr}\big[N_D\,\mathcal{L}_\alpha(\rho)\big], \tag{3}$$
$$J_{E,\alpha} = \text{Re Tr}\big[H_D\,\mathcal{L}_\alpha(\rho)\big], \tag{4}$$
$$J_{Q,\alpha} = J_{E,\alpha} - \mu_\alpha J_{N,\alpha}, \tag{5}$$

with $N_D = \text{diag}(0,1,1)$. Currents are cycle-averaged over the waiting interval and divided by T (leads off during the collision). We set

$$I \equiv \bar{J}_{N,L}, \qquad P_{\text{el}} \equiv -(\mu_L - \mu_R)I, \tag{6}$$

so $P_{\text{el}} > 0$ is electrical *output*. The battery channel is

$$P_{\text{erg}} = r\,\mathcal{W}_A, \qquad r = 1/T, \tag{7}$$

where $\mathcal{W}_A$ is the ergotropy of the outgoing ancilla. Continuous NESS checks satisfy $J_{N,L}+J_{N,R} \approx 0$ to numerical precision.

III. SPIN RESOURCE WITHOUT COLLISIONS

Figure 1 maps the continuous-lead steady state in the engine window of temperatures and chemical potentials used below. Spin coherence $|C| = |\rho_{\uparrow\downarrow}|$ is large at small Δ and intermediate tilt, and vanishes for collinear leads. Electrical power P_{el} is positive over a connected region of (Δ, θ_R). Thus the same nonequilibrium drive that supports thermoelectric operation also hosts a harvestable spin resource—without an engineered coherence target.

IV. THERMOELECTRIC ENGINE BRANCH

Fix temperatures $T_L > T_R$ and scan the load $V = \mu_L - \mu_R$. Figure 2 shows a clear engine branch at negative

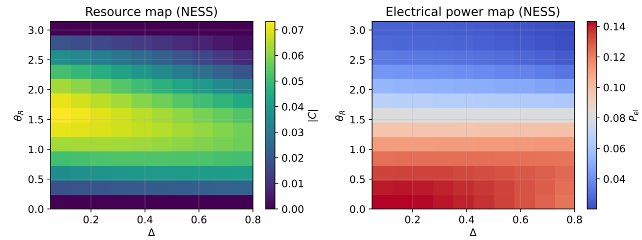

FIG. 1. NESS resource and electrical power versus level splitting Δ and right-lead tilt θ_R at fixed $(T_L, T_R, \mu_L, \mu_R, p)$. Left: $|C|$. Right: P_{el}.

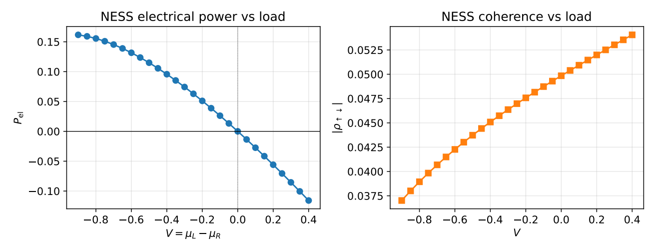

FIG. 2. NESS load sweep. Left: electrical output power versus $V = \mu_L - \mu_R$. Right: spin coherence. The engine branch is the region with $P_{\text{el}} > 0$.

V: the thermocurrent runs against the load, so $P_{\text{el}} > 0$, while $|C|$ remains finite. At the representative working point

$$\begin{aligned} &(T_L, T_R) = (2.0, 0.25), \quad (\mu_L, \mu_R) = (-0.3, 0.3), \\ &\varepsilon = 1.0,\ \Delta = 0.2,\ \theta_R = 0.6,\ p_L = p_R = 0.9, \end{aligned} \tag{8}$$

the continuous NESS has $P_{\text{el}} \approx 0.132$ and $|C| \approx 0.042$. A diagnostic efficiency $\eta \approx P_{\text{el}}/J_{Q,L}$ on the engine branch is reported only as a scaffold check, not as a Carnot-tight result.

V. COLLISIONAL BATTERY CHANNEL

We now add collisions at exchange angle $\theta \approx 0.45$ and scan the collision rate $r = 1/T$ on the engine window (Fig. 3). Electrical output stays positive for all rates in the scan, $P_{\text{el}} \in [0.097, 0.128]$. The battery power P_{erg} is smaller, of order 10^{-4}–10^{-5}, because the matched gap Δ sets the battery energy scale, but it is strictly positive for coherent exchange and peaks at higher r. Residual coherence remains of order a few percent. Thus the device keeps an engine branch while a second channel charges the ancilla tape.

VI. CONTROL HIERARCHY

Figure 5 compares protocols at fixed (T, θ). Coherent exchange produces $P_{\text{erg}} > 0$ with $\mathcal{W}_A$ purely coherent for

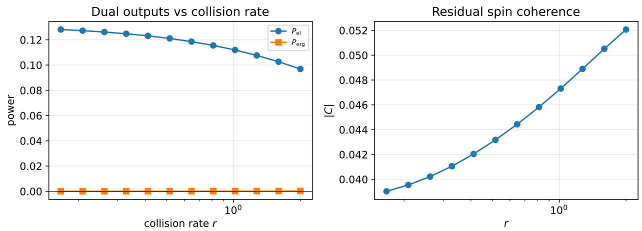

FIG. 3. Stroboscopic fixed point on the engine window. Left: dual powers versus collision rate. Right: residual spin coherence. P_{el} stays positive; P_{erg} opens only with collisions.

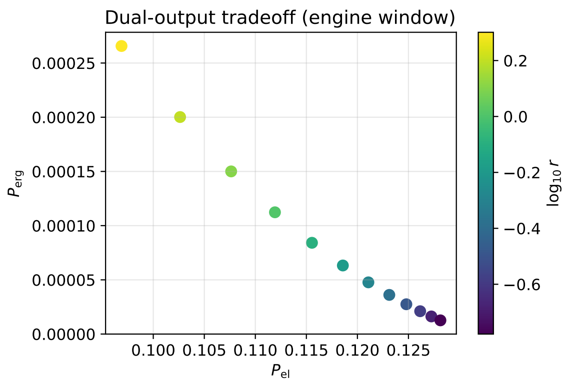

FIG. 4. Tradeoff scan $(P_{\mathrm{el}}, P_{\mathrm{erg}})$ along the rate path. Color encodes $\log_{10} r$.

ground-state ancillas. Incoherent exchange (spin dephasing before the unitary), no collision, collinear leads, pure dephasing, and $\alpha = 1/2$ all give $\mathcal{W}_{\mathrm{coh}} = 0$ (hence $P_{\mathrm{erg}} = 0$ or only population ergotropy). Matched-population ancillas strongly suppress the battery channel in this window. Collinear and dephasing protocols can slightly raise P_{el}, so local coherence is not required for electrical output and can even cost a small fraction of it. The battery channel, by contrast, requires coherent non-collinear exchange.

VII. COMPANION EXPORT LEMMA

The export step is the same matched-gap collision analyzed in a V-type companion model: an affine renewal of coherence between collisions yields an impedance-matching optimum $\Gamma_{\mathrm{ext}} = r(1-\cos\theta) \simeq \gamma_C$ for the charging power, with a flat small-angle plateau and a closed finite-precession factor. In the spin valve the "regrowth rate" is set by the leads rather than an engineered c_0. We therefore use the companion results as lemmas for the collision algebra and accessibility bookkeeping, not as a substitute for transport.

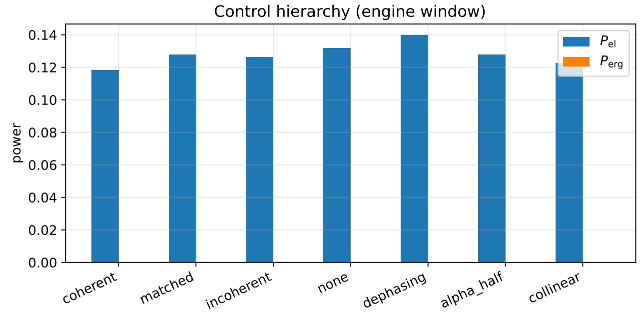

FIG. 5. Control hierarchy at fixed period and exchange angle on the engine window. Only coherent collisions open a clear battery channel.

VIII. DISCUSSION

The main story is dual output in a sequential-tunneling non-collinear valve: thermoelectric electrical power plus a collisional coherence battery. The battery channel is small compared with P_{el} at the present parameters, which is expected for a small matched gap. The honest reading is not that harvesting repairs the collinear engine, but that non-collinearity creates an additional resource that can be exported without destroying the engine branch.

Limits are explicit. Currents come from the sequential-tunneling generator, not NEGF. Ancilla preparation free energy is not subtracted. Coherent ergotropy of a single outgoing qubit is an upper bound without a phase reference [12]. Minimal thermal-qubit repeated-interaction no-go results [13] concern a different setup; they are cited only as background for collision models.

IX. CONCLUSION

A non-collinear Coulomb-blockade spin valve can operate as a thermoelectric engine while a tape of ancillas harvests its spin coherence through resonant exchange collisions. Lead-resolved, cycle-averaged currents keep $P_{\mathrm{el}} > 0$ on the chosen window for all collision rates we scanned. A control hierarchy attributes the battery channel to coherent exchange. The result is a dual-output picture of nonequilibrium spin-valve resources, with the collisional map controlled by a short companion analysis of impedance-matched harvesting.

ACKNOWLEDGMENTS

N.J. developed the numerical engine layer and the dual-output analysis. S.V. and B.M. proposed coupling a coherence-generating transport device to an ancilla tape. All authors contributed to the interpretation and the manuscript.

Appendix A: Lead generator and currents

For each lead α, Γ_α is diagonalized in spin space. Each eigenchannel with weight λ defines a spinor jump operator $L_+ = c_\uparrow|\uparrow\rangle\langle 0| + c_\downarrow|\downarrow\rangle\langle 0|$ with Fermi factors at the spinor energy. The full Liouvillian is $-i[H_D,\cdot]$ plus Lindblad dissipators for all channels. Particle and energy currents are $J_{N,\alpha} = \mathrm{Re}\ \mathrm{Tr}[N_D\mathcal{L}_\alpha(\rho)]$ and $J_{E,\alpha} = \mathrm{Re}\ \mathrm{Tr}[H_D\mathcal{L}_\alpha(\rho)]$. Cycle averages integrate these currents over the waiting interval and divide by T.

Appendix B: Collision channel

On the active block the exchange unitary is a partial iSWAP of angle θ. Tracing a diagonal ancilla reproduces the companion map for the doublet variables (s, d, C) in the singly occupied sector, with an occupation penalty from empty and doubly blocked weight outside that sector. At $\alpha = 1/2$ the local ancilla coherence vanishes and only population ergotropy can remain.

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